Diagnosing Structural Failures in LLM-Based Evidence Extraction for Meta-Analysis

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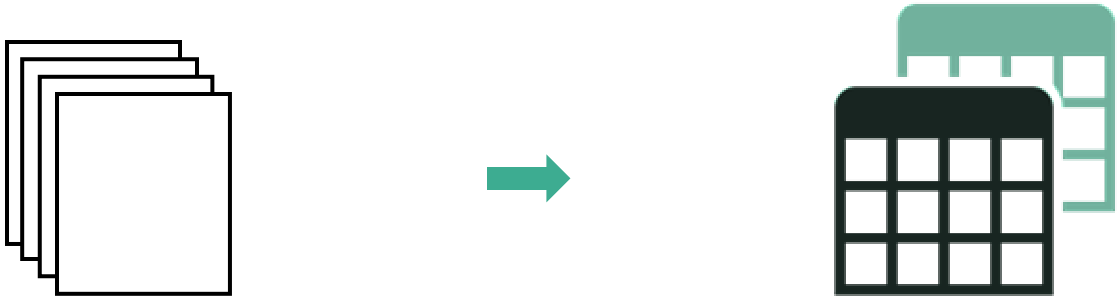
1. Why Meta-analysis?

Meta-analysis informs real-world decisions, from agricultural production and clinical treatment to environmental policy and social intervention.

These decisions depend on accurately extracting and aggregating statistically bound evidence from vast and heterogeneous scientific literature.

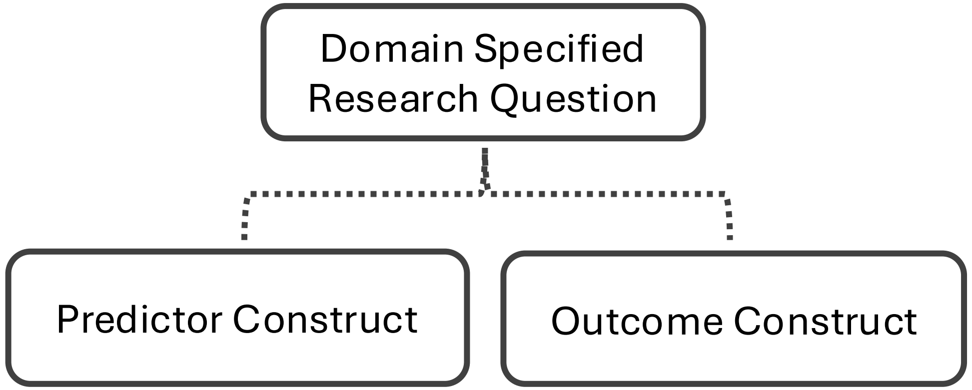
Collection of Empirical Studies

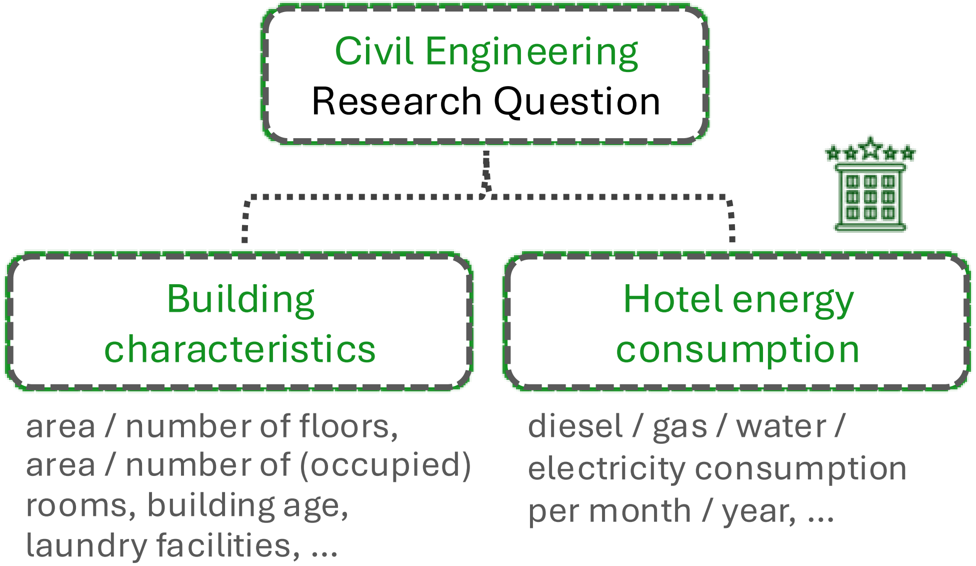


2. What is Meta-analysis?

For example,

- in civil engineering, we may aggregate evidence linking building characteristics to hotel energy consumption.

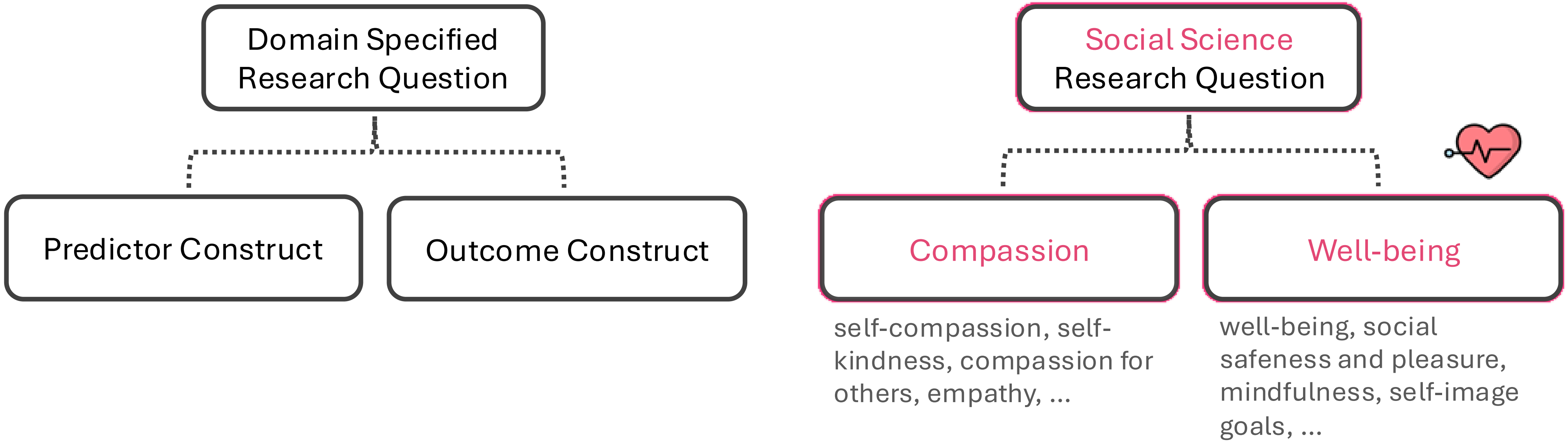




2. What is Meta-analysis?

For example,

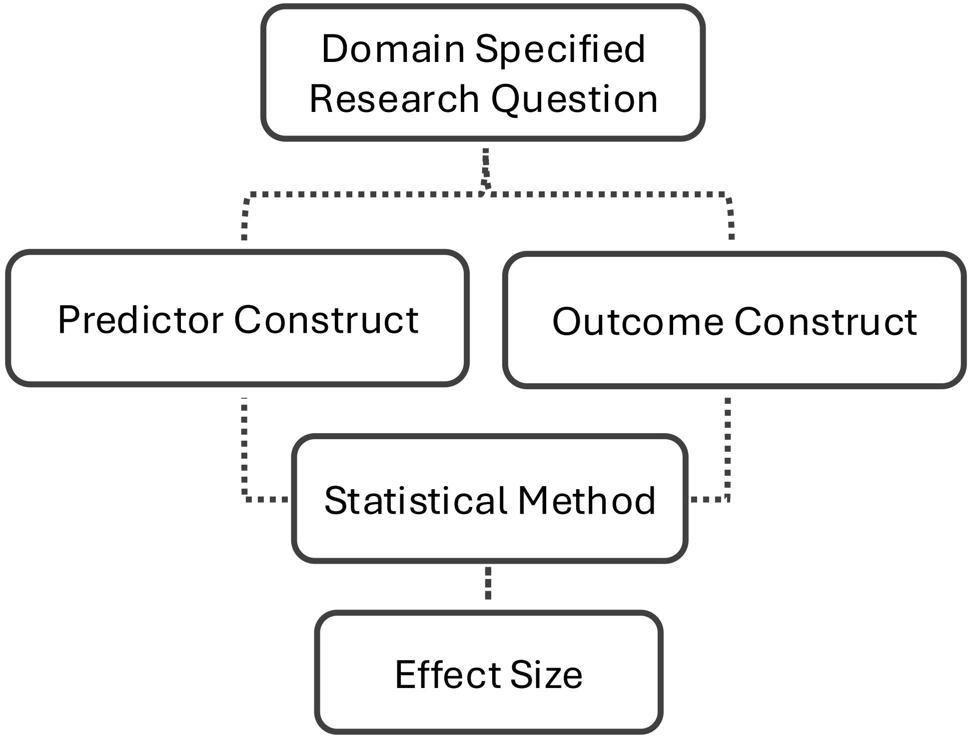
- in civil engineering, we may aggregate evidence linking building characteristics to hotel energy consumption.
- in social science, we may synthesize studies relating compassion to well-being.



2. What is Meta-analysis?

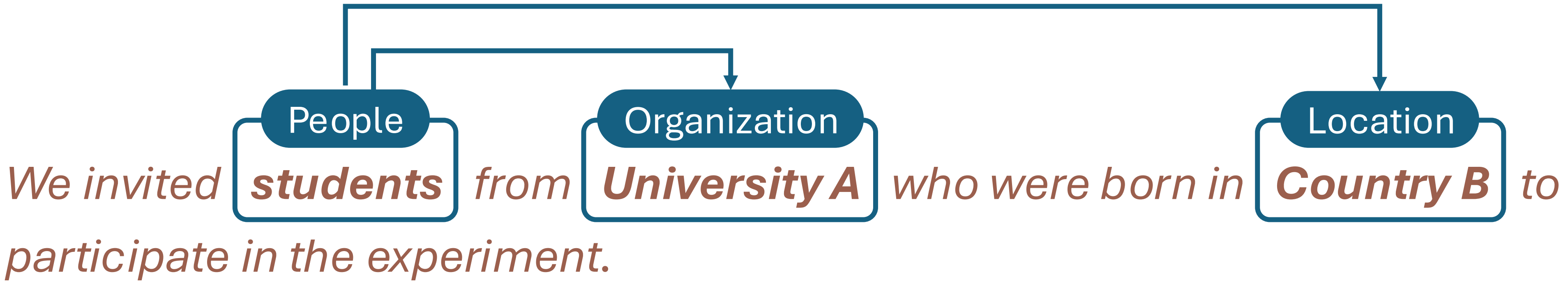
If these structural bindings are incorrect, downstream **conclusions** and the **policies built upon them** may become unreliable.

- a quantitative research synthesis method that aggregates effect size estimates derived from independent primary empirical studies.
- empirical studies report predictor–outcome associations together with their analytical models and effect size estimates.
- One empirical study **limited** by sample size, study population category, condition setting, etc.
- And before doing meta-analysis, you need to collect the this binding evidence across different empirical studies.



3. From Manual to Automated Extraction

- Traditionally, these structured associations were extracted manually by domain-specific researchers.
- The process is labor-intensive, slow, and difficult to scale as scientific output grows exponentially.
- With advances in natural language processing, researchers began exploring automation: TF-IDF, named entity recognition, relation extraction, sequence labeling, LLM-based document QA



4. Challenges of Automated Extraction

However, scientific evidence extraction introduces additional complexity. It becomes a **multi-modal, multi-hop, multi-document**, and **structurally constrained reasoning task**.

- Information may appear in tables, figures, or appendices

- Answer often require multi-step reasoning across multiple textual nodes, identifying the relevant elements and binding them correctly is non-trivial

- Scientific documents are typically long and structurally dense, and complexity increases further when multiple documents are processed jointly.

- Numeric values must be accurately grounded and correctly attributed to the corresponding variables, models, and conditions.

5. Method: Structured Define

Property	Definition	Notation
Document ID	Index of the source document in the corpus	$i \in \{1, \ldots, \lvert\mathcal{D}\rvert\}$
Study Population	Study population or unit of analysis	$P \in \mathcal{P}$
Geolocation	Country of study conduct (normalize cities/regions to countries)	$G \in \mathcal{G}$
Sample Size	Main reported sample size for the study	$N \in \mathcal{N}$
Statistical Method	Statistical or association method (e.g., Pearson, regression, OR)	$A \in \mathcal{A}$
Variable	Variables including predictor variable & outcome variable	$V \in \mathcal{V}$
Scale and Unit	Atomic pair describing measurement scale and unit	$(S, U) \in \mathcal{SU}$
Conditions	Optional contextual qualifier for an association estimate	$C \in \mathcal{C}$
Effect Size	Reported association estimate (e.g., r, β, OR, etc.)	$E \in \mathcal{E}$

5. Method: Structured Define – one element

Object-related **1**

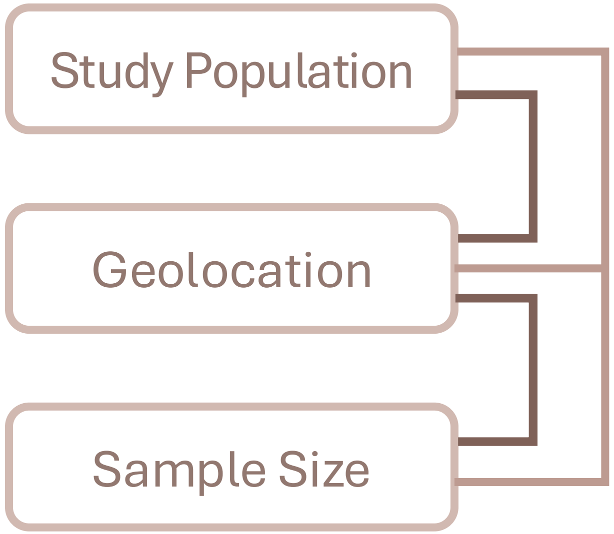
- Study Population
- Geolocation
- Sample Size

Method-related **1**

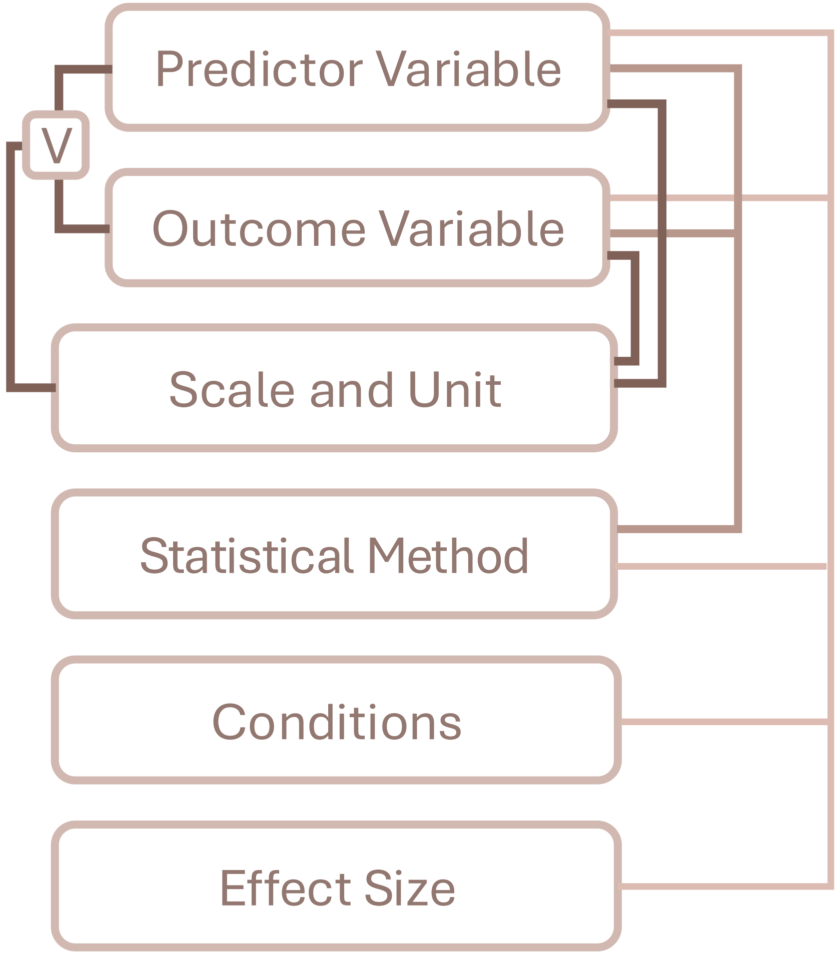
V
- Predictor Variable
- Outcome Variable
- Scale and Unit
- Statistical Method
- Conditions
- Effect Size

5. Method: Structured Define – bonded elements

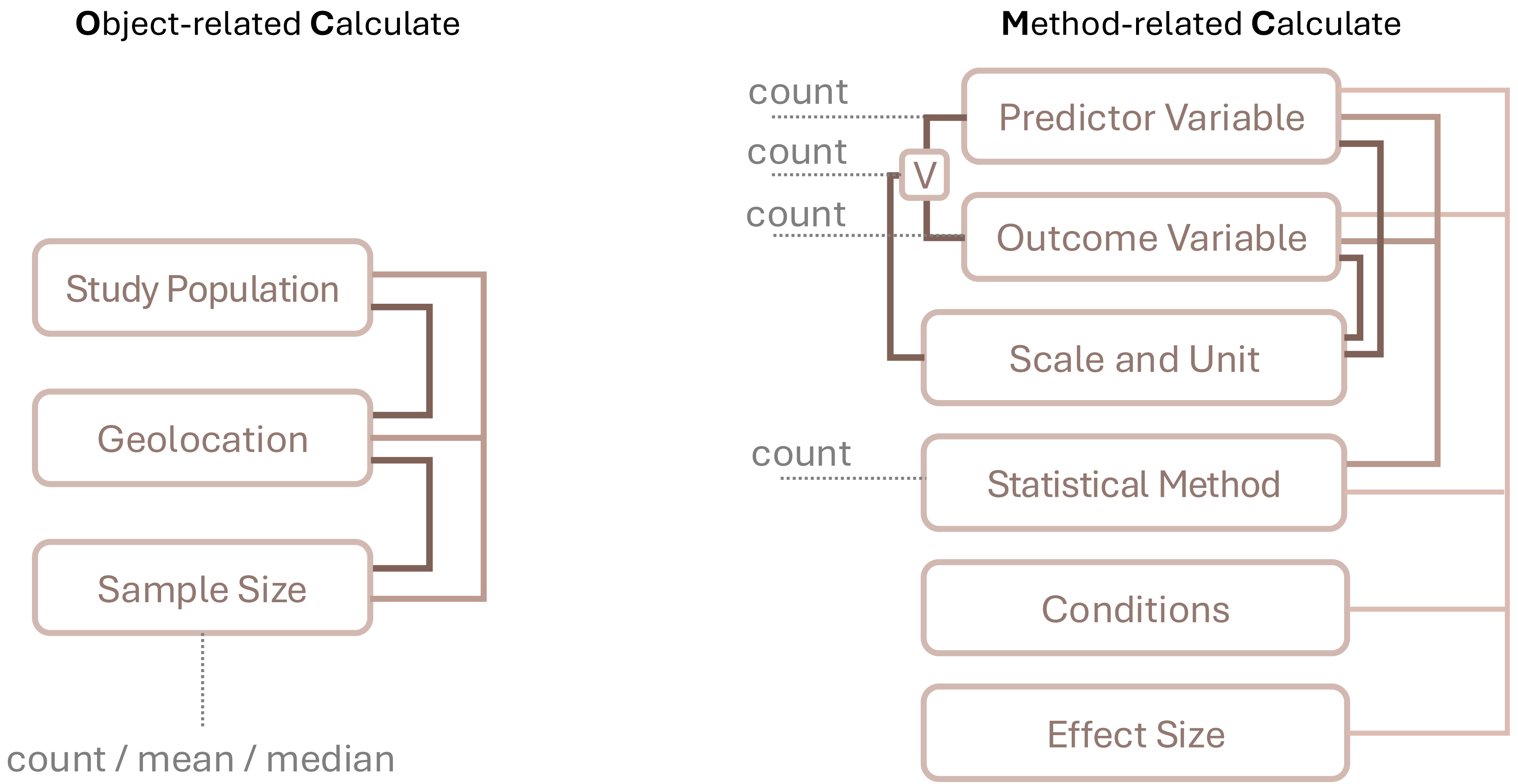
Object-related **2**



Method-related **2**



5. Method: Structured Define - calculating



5. Method: Dataset Introduction

Five domains

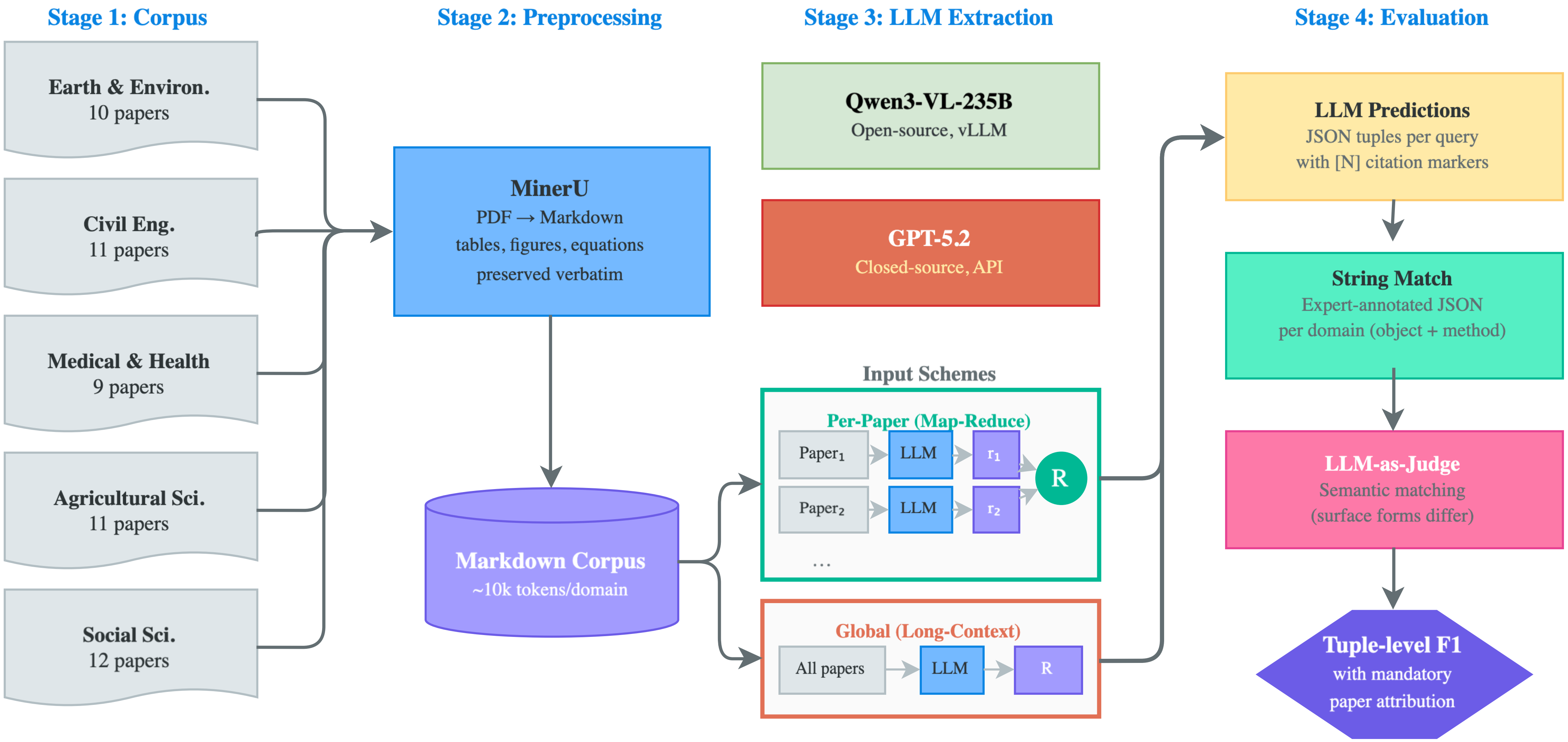
Domains	Research Questions	Source
Civil Engineering	Building characteristics → Hotel energy use	A review paper
Medical & Health	Anthropometric indicators → Ophthalmic outcomes	A review paper
Earth & Environment	Carbon storage → Biodiversity metrics	A review paper
Agricultural	Water stress → Plant responses	Search in a Journal
Social	Compassion → Well-being	A review paper

5. Method: Dataset Statistics

Count: Sum / Min / Max

Domains	Paper	Statistical Method	Variables	Effect Size
Civil Engineering	11	23 / 1 / 5	139 / 2 / 32	258 / 2 / 46
Medical & Health	9	18 / 1 / 3	71 / 2 / 16	328 / 16 / 78
Earth & Environment	10	12 / 1 / 2	84 / 4 / 19	198 / 3 / 35
Agricultural	11	20 / 1 / 6	72 / 4 / 11	226 / 4 / 54
Social	11	15 / 1 / 2	81 / 2 / 18	137 / 2 / 28

5. Method: LLM Meta Analysis Pipeline



6. Result – Marco F1

Global

Model	**O1**	**O2**	**OC**	**M1**	**M2**	**MC**	**All**
Qwen3-VL	0.57	0.27	0.2	0.21	0.05	0	0.18
GPT-5.2	0.66	0.18	0.07	0.39	0.2	0.07	0.24
Avg.	0.62	0.23	0.14	0.30	0.13	0.04	0.21

Object-related (O1, O2, OC) tasks achieve higher F1 scores than **method-related tasks** (M1, M2, MC).

Tasks involving **bonded elements** (O2, M2) perform substantially worse than **single-element tasks** (O1, M1).

6. Result

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Per-Paper

Model	**O1**	**O2**	**OC**	**M1**	**M2**	**MC**	**All**
Qwen3-VL	0.62	0.3	0.33	0.45	0.22	0.28	0.35
GPT-5.2	0.5	0.2	0.27	0.45	0.22	0.19	0.28
Avg.	0.56	0.25	0.30	0.45	0.22	0.24	0.32

Per-paper extraction improves overall performance by an average of 11% (Qwen3-VL: +17%, GPT-5.2: +4%).

The improvement is particularly pronounced for **bonded-element tasks (M2)** and both **calculation-related tasks** (OC, MC).

This suggests that breaking down long-context inputs into per-document processing helps stabilize structural binding and improves performance.

6. Error Analysis: **The LLM Knows *What*, But Not *Which Is Which***

The LLM correctly identifies both variables in a causal relationship but **swaps their roles**.
It puts the cause where the effect should be, and vice versa.
Swaps appear in **4 of 5 domains** (Social 176, Earth 152, Medical 147, Agricultural 5)

Example: Biodiversity Domain

In these papers, ecosystem services (water regulation, soil retention) predict the **presence** of species not the other way around. But the LLM assumes species cause ecosystem outcomes:

	Predictor Variable	→	Outcome Variable
LLM	Birds	→	Water flow regulation
Ground Truth	Water flow regulation	→	Birds
LLM	Frogs	→	Soil accumulation
Ground Truth	Soil accumulation	→	Frogs

Acknowledgment

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Any comments or questions?

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GitHub: https://github.com/zhiyintan/LLM-Meta-Analysis